

120 Days of Quantitative Finance

Month 4: Stochastic Calculus

Week 13: Brownian Motion

Day 88: Ito's Integral and Ito's Isometry

1 Introduction

In classical calculus, we integrate with respect to time:

$$\int f(t) dt$$

In stochastic calculus, we integrate with respect to Brownian Motion:

$$\int f(t) dW_t$$

This is called the **Ito integral**.

2 Definition of Ito Integral

For a process $f(t)$, the Ito integral is defined as:

$$\int_0^T f(t) dW_t = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} f(t_i) (W_{t_{i+1}} - W_{t_i})$$

Key difference from classical integrals:

- The integrand is evaluated at the **left endpoint**
- Brownian increments are random

3 Key Intuition

- dW_t is random noise
- The integral accumulates weighted randomness
- The result is itself a random variable

4 Ito Isometry

One of the most important results:

$$E \left[\left(\int_0^T f(t) dW_t \right)^2 \right] = E \left[\int_0^T f^2(t) dt \right]$$

4.1 Interpretation

- The variance of the stochastic integral equals the integral of squared function
- This is analogous to energy conservation

5 Special Case

If $f(t) = 1$:

$$\int_0^T dW_t = W_T$$

and:

$$E[W_T^2] = T$$

which matches Ito Isometry:

$$E \left[\left(\int_0^T 1 \cdot dW_t \right)^2 \right] = \int_0^T 1^2 dt = T$$

6 Why Ito Integral Matters

Ito integrals are fundamental for:

- Stochastic Differential Equations (SDEs)
- Ito's Lemma
- Financial modeling (Black-Scholes)

7 Conclusion

Ito's Integral extends classical integration to stochastic processes.

Ito's Isometry provides a powerful tool for analyzing variance and stability.

These concepts are essential for working with continuous-time financial models.